

SANGUINE

OS17A - Project Baseline & Tradespace Exploration

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Sanguine is a physician-facing clinical interpretation layer for Mayo Clinic laboratory results. It aims to improve Mayo Clinic lab patient care outcomes at scale by augmenting physician-facing test results with relevant research using an agentic AI clinical librarian.

1. Risk Identification and Mitigation

Modeling Risk

As a team building a model from scratch, we had a design choice about how to model risks. We first implemented rich risk modeling. Risks could add arbitrary tasks that could block other tasks, and every task had a rework risk. Each risk had a function defining whether it would trigger and another function defining when it'd be discovered, and risks triggering other risks. *All models are wrong, but some models are useful.* We realized that in our attempt to make our risk model less wrong, we'd also made it less useful. Requiring every risk to define two functions and the many fields of a Task made the model more difficult to understand, and even visually, point clouds for each project would be a less clear way to represent riskiness than color. In this scenario, adding complexity through Monte Carlo stochastic logic detracted from the model's usefulness. The ability to clearly review model exposure to risk and the Pareto frontier of risk versus cost benefit from the clarity of simplicity.

We simplified our modeling framework (the model link is in *Appendix 3*). Now, we only model two fields per risk: effect and probability. Effect is graded qualitatively on a 1-100 scale, where 100 is a catastrophic impact that would surely result in cancellation. A risk's total impact is its effect times its probability. We model a project's riskiness as the sum of its impact of each risk. Impact is a surrogate for risk exposure. This does not capture interactions between risks, but we exclude them because we found the interactions between risks to create complexity and clouded that ability to draw insightful conclusions within our previous model.

Risk Identification

This risk log defines the main project risks used in the OS17 mitigation analysis. Each risk was identified from the expected failure modes of the Sanguine project architecture, especially the dependencies among data integration, agentic AI development, clinical validation, regulatory approval, compliance, and physician-facing deployment. The table assigns each risk an estimated effect with a range of 1 to 100 and a probability range of 1 to 100%. The impact score is calculated as Effect x Probability, allowing the risks to be compared using a common scale.

ID	Risk	Effect	Probability	Impact	Commentary
R1	(EHR) / Lab Integration Failure	70	50%	35	Mayo Clinic's Laboratory Information Management System (LIMS) and EHR systems may not expose lab data via accessible APIs or in expected formats. Data governance approvals are uncertain and could block Data Interface & Harmonization and all downstream tasks.
R2	AI Clinical Accuracy Failure	85	40%	34	The agentic AI librarian may generate factually incorrect clinical interpretations, creating patient-safety liability and regulatory barriers to Shadow Mode & Clinical Review and deployment.
R3	(IRB) / Regulatory Delay	60	55%	24	Clinical AI tools require IRB review and may require FDA oversight. Approval timelines are unpredictable and could block Shadow Mode for months, delaying the entire pilot and global launch.
R4	Patient Data Compliance Failure	90	15%	13.5	Handling patient lab data requires strict HIPAA compliance. A breach or regulatory audit finding could halt the project entirely. Effect is critical but probability is low given Mayo Clinic's existing compliance infrastructure.
R5	Clinical Validation Bottleneck	65	35%	26	Clinical validation depends on Mayo specialists and physician reviewers being available to evaluate outputs and confirm if the system is clinically useful. Even if the technical prototype works, limited reviewer capacity could slow Shadow Mode & Clinical Review, reduce the validation set, or even delay the pilot decision.
R6	Late Interface Rework	75	40%	30	The work related to the data pipeline, retrieval layer, agentic AI core, physician-facing interface, and validation workflow may work independently but may fail when integrated. If unexpected interface premises or specific needs are only discovered late, it may require

					cascading rework in the work of Data Interface & Harmonization, Data Context & Retrieval Pipeline, Agentic AI Core Development, and Shadow Mode & Clinical Review, affecting technical confidence and increasing residual risk.
R7	Team Knowledge Loss	55	30%	16.5	The project depends on knowledge shared across AI, Data, Cloud, QA, and Clinical Workflow teams. If key team members leave during development, or if contracted members' turnover/handoffs to other contractors are not well managed, then the teams may lose context for important parts of the project. This is especially true regarding interface assumptions and compliance constraints. This could delay Agentic AI Core Development, Data Interface & Harmonization, Regulatory & Compliance Readiness, and Shadow Mode & Clinical Review, while increasing rework across these phases.

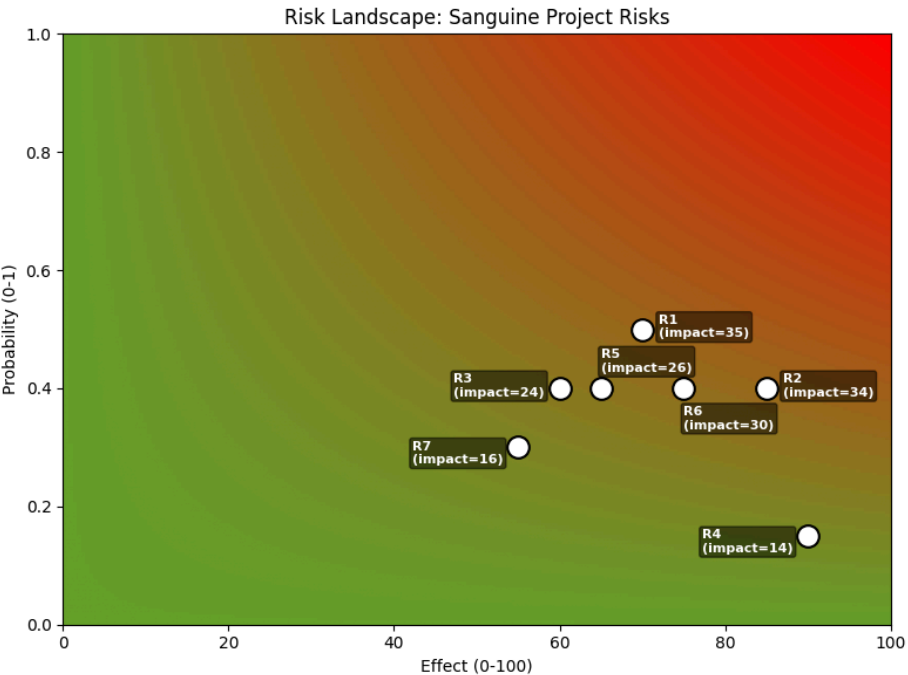


Fig 1.1 - Risk landscape plot

highly probable under the assigned probability in the risk log (55%). R6 (Late Interface Misalignment / Cascading Rework) is also critical because late misalignments can cause rework across data pipeline, retrieval layer, agentic AI core, and clinical validation workflow phases. R4 (HIPAA / Patient Data Compliance Failure) has the largest negative effect but a lower probability, as we don't expect blocks due to compliance data use. For this reason, it remains important but less central to mitigation decisions. R5 (Clinical Validation Resource Bottleneck) and R7 (Team Turnover / Knowledge Loss During Development) are relevant execution risks but are below the main exposure cluster.

Mitigation Activities

We define mitigations for the main risk areas identified in the risk log: R1 (EHR / Lab Data Integration Failure), R2 (AI Clinical Hallucination / Accuracy Failure), R3 (IRB / Regulatory Approval Delay), R4 (HIPAA / Patient Data Compliance Failure), and R6 (Late Interface Misalignment / Cascading Rework). Each mitigation is modeled as a project task, adding labor effort, non-labor

The matrix represents each risk by Effect and Probability, with the background gradient indicating increasing risk exposure from green (lowest risk, 0 of impact) to red (highest risk, 100 of impact). Based on our risk log, R1 (Electronic Health Record (EHR) / Lab Data Integration Failure), R2 (AI Clinical Hallucination / Accuracy Failure), R3 (Institutional Review Board (IRB) / Regulatory Approval Delay), and R6 (Late Interface Misalignment / Cascading Rework) form the highest risk exposure cluster, as they combine a higher probability and effect in the project. R1 (EHR / Lab Data Integration Failure) and R2 (AI Clinical Hallucination / Accuracy Failure) have the highest impact scores because failures in data integration and clinical accuracy can directly block development and subsequent phases. R3 (IRB / Regulatory Approval Delay) is important because regulatory delays are

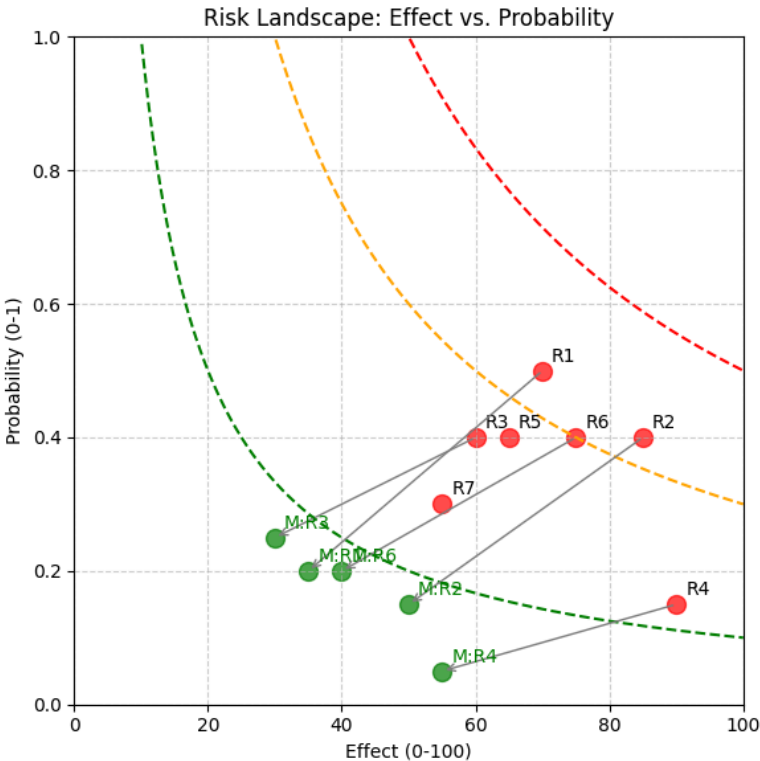


Fig 1.2 - Risk mitigation landscape review

cost, and schedule dependencies. Together, these mitigations move the selected risks out of the highest-impact region and into a lower-impact region by reducing their assigned effect, probability, or both. The table below, in accompaniment with the table found in *Appendix 1*, describes how each mitigation reduces the original risk effect and probability, while the following section compares the resulting project designs in terms of cost, schedule, and residual risk. The following risks, R5 (Clinical Validation Resource Bottleneck) and R7 (Team Turnover / Knowledge Loss During Development), are accepted risks in this design iteration. The risk R5 is accepted because clinical reviewer capacity depends primarily on Mayo specialist availability, which the project team cannot fully control. The risk R7 is accepted because turnover and handoff risks are managed through baseline project management, documentation, and team coordination rather than through a separate mitigation task. Both risks remain in the residual risk model.

Project Designs

Each mitigation is treated as an incremental project design in the design walk. Adding a mitigation task reduces the project's total residual risk and provides key benefits, but it also changes costs and, in some cases, the schedule. Starting with interface alignment mitigation (M6), it adds data integration feasibility (M1), clinical validation planning (M2), regulatory readiness (M3), and PHI compliance review (M4). This sequence is able to reduce total risk exposure by 58%, from 179 in the baseline to 75 in the full mitigation design. However, the reduction is not free: the estimated cost increases by 47% from \$4,798,352 to \$7,095,600, reaching the budget limit, and the schedule increases by 8% from 46 to 50 weeks, also reaching the schedule limit.

Design Name	Mitigations	Cost	Schedule (weeks)	Riskiness	Commentary
D0 No Mitigations (Baseline)	None	\$4,798,352	46	179	Unmitigated baseline. All risks at full exposure.
D1 Interface Alignment	M6	\$5,322,664	47	157	Adds interface review and early integration testing. This first mitigation reduces late interface misalignment and cascading rework risk before integration problems propagate across the data pipeline, retrieval layer, agentic AI core, and validation workflow.
D2 Data Integration	M6, M1	\$5,542,664	47	129	Adds data integration feasibility spike. This design reduces both late interface misalignment and EHR/Lab data integration risk, creating a stronger technical feasibility baseline before clinical validation is introduced.
D3 Clinical Validation	M6, M1, M2	\$6,451,288	49	102	Adds clinical validation protocol and physician expert panel. This design reduces technical feasibility and clinical accuracy risk, but it crosses the \$6M budget threshold because clinical validation adds significant review effort.
D4 Regulatory Roadmap	M6, M1, M2, M3	\$6,731,288	49	86	Adds IRB pre-submission and a regulatory roadmap. This reduces regulatory approval delay risk after the technical and clinical validation path has already been partially de-risked.
D5 Full Mitigation	M6, M1, M2, M3, M4	\$7,095,600	50	75	Completes the incremental mitigation path by adding PHI handling and compliance design review. This option produces the lowest residual risk, effectively mitigating hallucination and regulatory blockers, but it also has the highest cost and reaches the 50-week schedule limit.

2. Visualize the Project Tradespace

Tradespace Time vs Cost

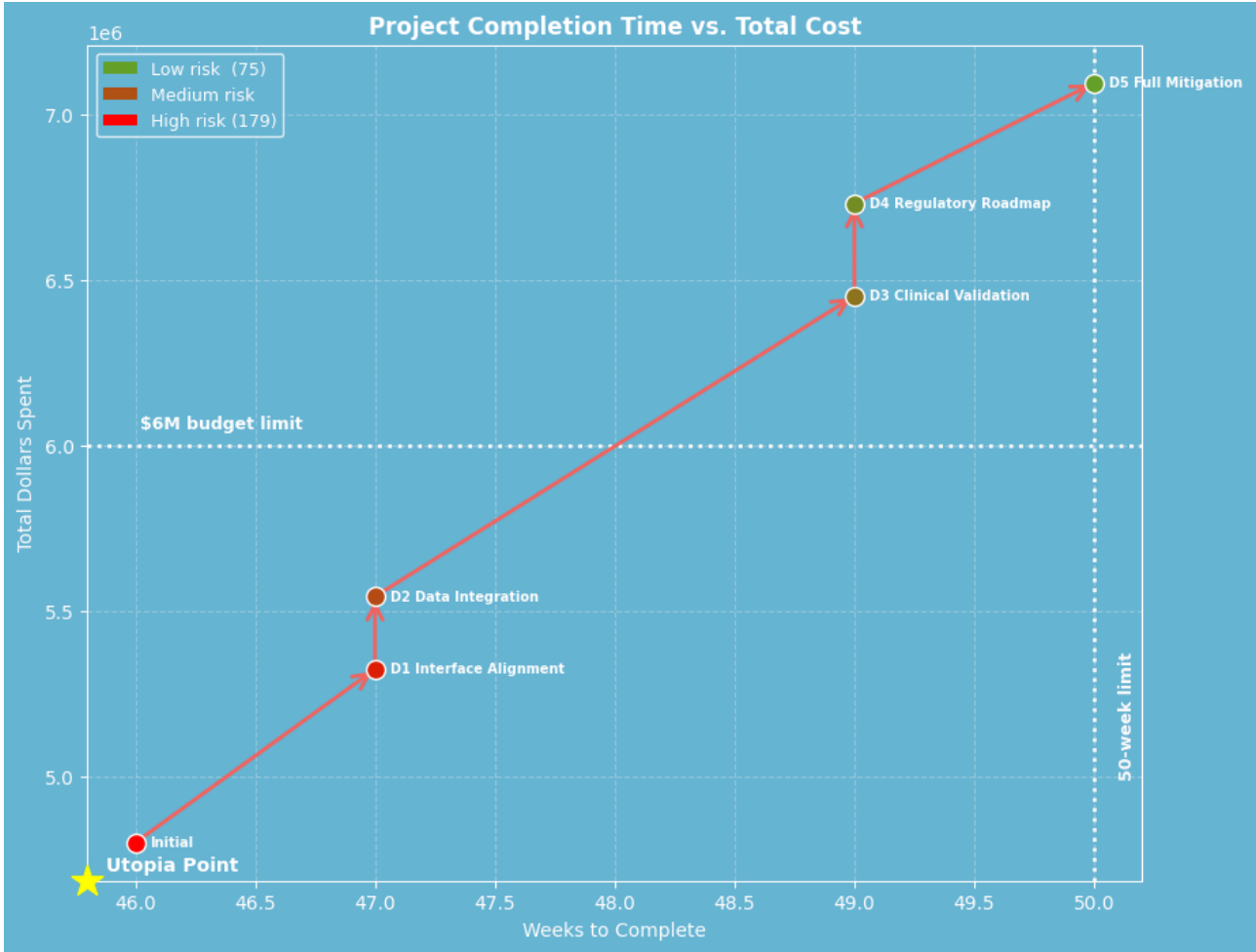


Fig 2.1 - Tradespace of schedule versus cost for design decisions

Tradespace Cost vs Riskiness

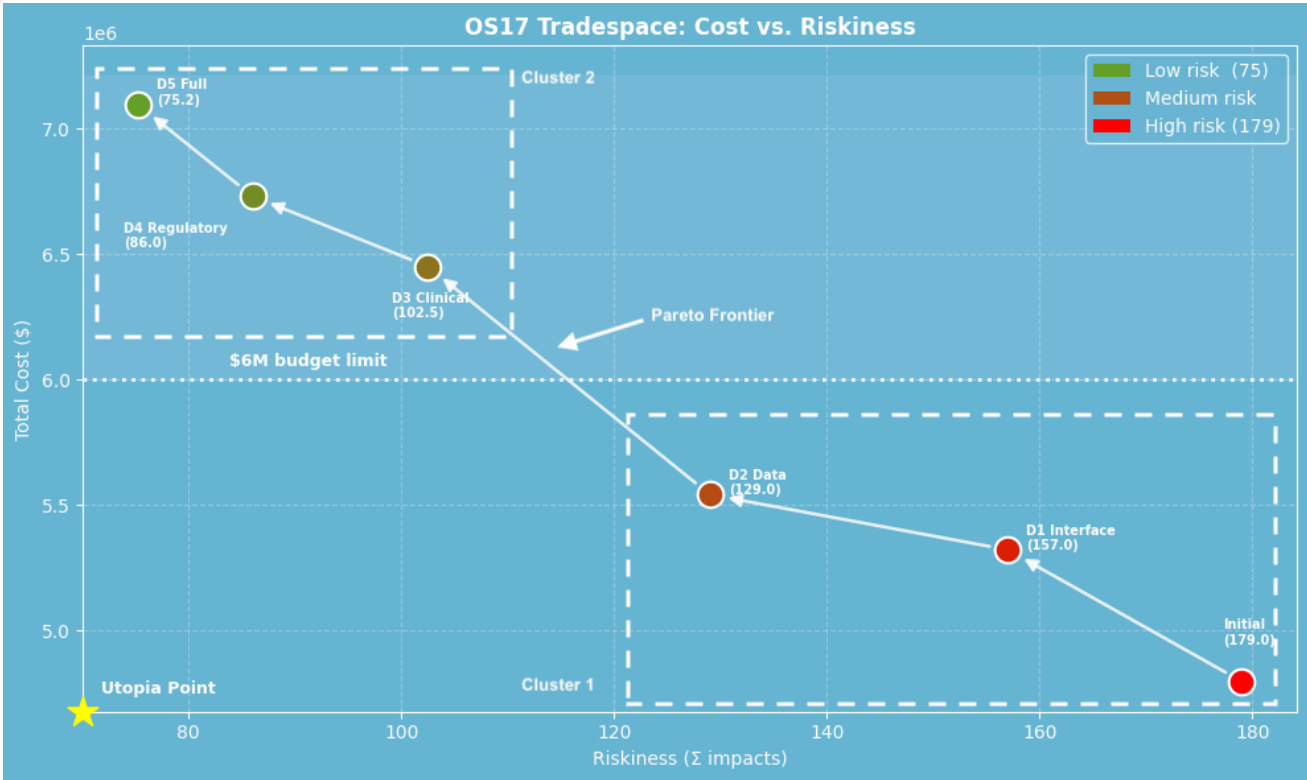


Fig 2.2 - Tradespace of risk versus cost for design decisions

Introduction:

The design walk starts with the unmitigated baseline and progressively adds mitigation work packages to reduce the most significant project risks. The path shows how the project progresses as more front-end risk reduction is added, starting with interface alignment, then data integration, clinical validation readiness, regulatory readiness, and finally, PHI compliance control.

The target region, from OS16, is defined by the project's final budget and schedule limits: \$6M and 50 weeks. Thus, this is the preferred planning envelope for the project. Designs that exceed the budget and schedule limits are still informative, but they represent greater project exposure.

In both tradespace views, marker color represents residual riskiness. The riskiness score is normalized within the observed design-walk range, from the baseline value of 179 to the full mitigation value of 75.2. The normalized value is calculated as $(\text{riskiness} - 75.2) / (179 - 75.2) \times 100$. Values closer to 100 are shown as higher risk, values around the middle of the range as medium risk, and values closer to 0 as lower risk. Therefore, the colors compare the designs within this design walk, using the baseline as the highest-risk point and the full mitigation design as the lowest-risk point.

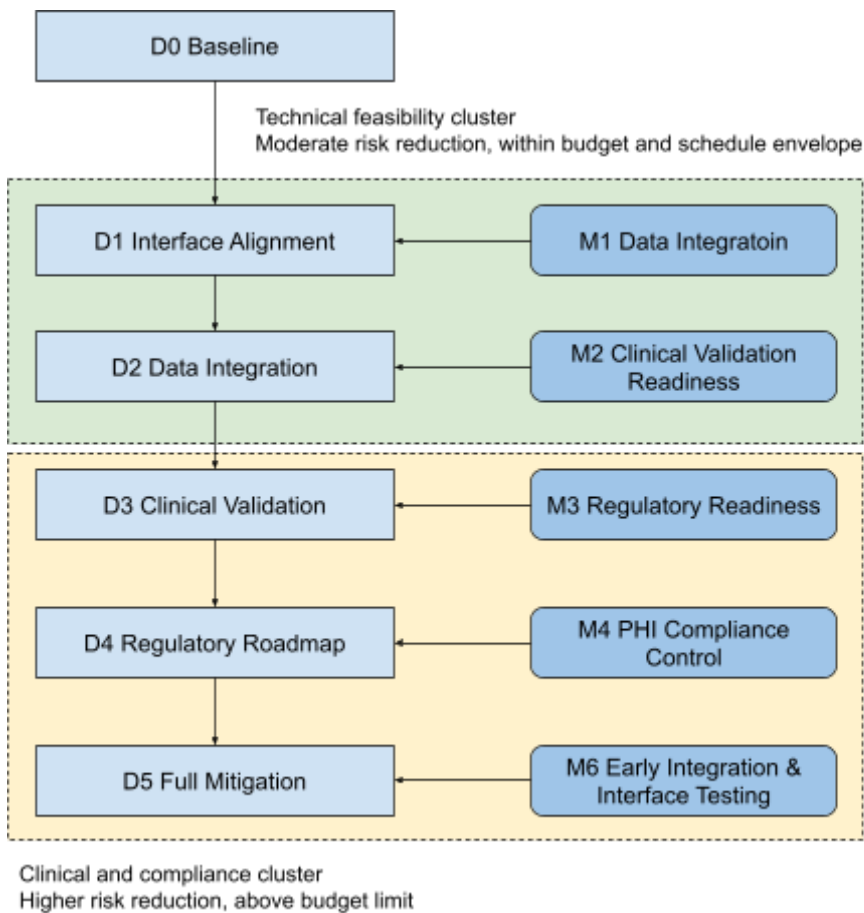


Fig 2.3 - Clustering of designs and associated mitigation

while D5 (Full Mitigation) is the full mitigation design, adding PHI compliance controls and reaching the 50-week schedule limit with an extra cost of approximately \$1,095M above the budget limit, corresponding to a close to 20% of increase. This shows that the highest impact on cost and schedule results from clinical validation, regulatory readiness, and compliance mitigation, rather than from technical mitigation measures. From this perspective, project D2 is the most attractive within the preferred planning scope, while projects D3 to D5 represent more robust mitigation options that require significant budgetary investment.

Second Tradespace:

This tradespace focuses on the comparison between risk and cost, demonstrating a gradual reduction of risk from the baseline, starting at 179 and reaching the final design at 75, indicating a reduction of more than 50% along the

First Tradespace:

The first tradespace (schedule x cost) shows the consequences of the incremental mitigation path, composed of 5 stages. Each of these stages represents a new design that is found upon the Pareto frontier. Risk exposure is incorporated as a third dimension through the color of the markers, allowing each design to be analyzed also by residual risk. In the first walk, with D1 (Interface Alignment) and D2 (Data Integration), both within the preferred planning envelope, it is indicated that initial technical mitigations can be adopted without exceeding the established budget or schedule limits. The main architectural shift occurs at D3 (Clinical Validation). Integration of Clinical Validation forces the project \$451K outside the \$6M target threshold and extends the schedule from 47 to 49 weeks. This demonstrates the high burden of validation within a clinical setting. D4 adds Regulatory Roadmap at an additional cost of approximately \$731K above the budget limit with no additional schedule, while D5 (Full Mitigation) is the full mitigation design, adding PHI compliance controls and reaching the 50-week schedule limit with an extra cost of approximately \$1,095M above the budget limit, corresponding to a close to 20% of increase.

way. Despite the reduction, each risk reduction implies an additional cost that can reach up to an 18% increase in the projected budget. Starting with the first region within the budget limit, D1 (Interface Alignment) and D2 (Data Integration) designs offer the most attractive initial walk, as they reduce riskiness from 179 to 129, a reduction of approximately 30%. The second region, above the budget limit, D3 (Clinical Validation) and D4 (Regulatory Roadmap) add clinical validation and further reduces regulatory readiness, reaching a residual risk of 86. In contrast, D5 (Full Mitigation) reaches the lowest risk of 75.2 through the full mitigation design.

The second tradespace supports the same decision as the cost–schedule: D2 (Data Integration) is the strongest design within the budget cost target, while D3 (Clinical Validation) to D5 (Full Mitigation) are stronger risk-reduction designs that require accepting additional budget exposure.

Clustering:

The design walk of both tradespaces reveals 2 clear clusters. The first one, composed of D1 (Interface Alignment) and D2 (Data Integration), focuses on technical feasibility, aiming to reduce interface and data integration uncertainty while remaining within budget and schedule limits. The second cluster, composed of D3 (Clinical Validation) and D5, forms the clinical and regulatory readiness. Despite the second cluster providing stronger risk reduction, it also pushes the project above the budget limit. Figure 2.3 illustrates these design clusters and associated mitigations.

Rework Risk:

Rework is represented in the model as an explicit project risk, rather than a complete and separate rework task. Risk R6, Late Interface Misalignment / Cascading Rework, captures the possibility that the data pipeline, retrieval layer, agentic AI core, physician-facing workflow, and validation workflow will function independently but fail when integrated. In response, mitigation M6 (Early Integration & Interface Testing) is positioned early in the project planning phase to reduce this risk before the project reaches shadow-mode clinical validation, where rework would be more expensive and more likely to propagate across phases.

Coordination and communication:

Coordination and communication efforts are captured by the model's mitigation work and productivity assumptions. The most comprehensive mitigations are M2 (Clinical Validation) and M6 (Early Integration & Interface Testing), as they require coordination across multiple cross-functional teams: AI, QA, PM, SWE, DATA, CLOUD, clinical reviewers, and compliance support. Their work estimates represent the effort distributed across all teams, not the duration of a single team. This is important to mention because the main risks do not stem from isolated technical tasks. They emerge from dependencies between teams, data integration, clinical validation, and governance. The cost increase between D2 to D5 is driven in part by increased coordination overhead.

Evaluation Summarized:

The design walk analysis also demonstrates how front-end work reduces uncertainty before construction and testing. M1 (Data Integration) uncovers blockers in EHR/Laboratory data through integration, M6 (Early Integration & Interface Testing) exposes interface assumptions through initial integration testing, M2 prepares for clinical validation before shadow mode, M3 (Regulatory Readiness) and M4 (PHI Compliance Review) reduce the likelihood of discovering regulatory or PHI compliance gaps only at the validation stage. Error propagation is minimized by testing interfaces earlier, clarifying validation criteria before clinical review, and treating compliance as a control constraint before patient data is used. If the project deviates significantly from the original plan, design walk analysis helps the team react with different actions: technical problems justify expanding the plan to include D1 (Interface Alignment) and D2 (Data Integration), clinical or regulatory problems justify adding D3 (Clinical Validation) and D4 (Regulatory Roadmap), and compliance findings may require complete mitigation, represented by D5. Each change should be evaluated for its impact on cost, schedule, and residual risk.

3. Preferred Options and Baseline

Executive Summary:

- We have transitioned from our proposal from the previous plan titled “Strategic Scaling” to a heavily risk mitigated plan “D5 Full mitigation”.
- This transitions from a plan with a 46 week timeline and budget of \$4.69M (previous), to a plan with a 50 week timeline and costs of \$7.10M (suggested new baseline).
- This exceeds the previously proposed budget and leaves no slack in the previously proposed schedule. We believe these are necessary and articulate our justification in more detail below.

Why cost increases are necessary: In the past we had highlighted key risks within the project as cascading rework, and clinical hallucinations as problematic areas that could derail Sanguine during development and deployment. Our review of risks within our proposal and the cost of mitigation seem like a steep departure from our previous baseline, with a cost increase of \$2.4M. If we believe in a high probability of these risks being realized, then investment at the outset of the project to mitigate is prudent. As a project progresses the cost to address latent problems increases non-linearly. Front loading these costs curtails the issue of cascading rework to manifest. Investment in the key risk mitigations of “Data Integration Spike” and “Clinical Validation.” The design walk reveals no classical Pareto frontier in cost × schedule space — every mitigation either increases both cost and schedule, or cost alone. D5 is not Pareto-optimal in cost × schedule terms, but it is the dominant choice when riskiness is treated as the third dimension. D1 and D2 represent the most cost-efficient mitigations, remaining within the \$6M budget. The critical transition occurs at D3, where clinical validation adds \$900K and 2 weeks — a steep but unavoidable step given patient safety requirements. D4 adds regulatory readiness at low marginal cost with no schedule impact; D5 adds PHI compliance controls, reaching the 50-week limit

Why sustaining risk is a non option: A key component that is assumed but must be explicitly made clear is that patient outcome risks introduced by Sanguine are non options for Mayo Clinic. Brand trust in the Mayo Clinic Model of Care is paramount to defend in the process of introducing Sanguine. Compromising data integrity or risking data breaches are extremely costly to the carefully cultivated brand trust that Mayo Clinic holds. Sanguine's original concept is to improve patient care outcomes by augmenting physician-facing lab results with relevant research using an agentic AI clinical librarian. D5 directly protects this value — a system that hallucinates clinical interpretations or fails to integrate with Mayo's EHR does not augment physician judgment, it undermines it.

Residual risk: These mitigation efforts still contain some latent non-addressed risk exposure but are vastly reduced relative to the previous “Strategic Scaling” project recommendation. This failure to properly address these risks in our past proposal, instead opting to minimize schedule time and cost requires a realization of that risk to not occur to become feasible. This is unlikely and instead a plan that incorporates recognition of risk can be viewed as a new fully developed plan, rather than ballooning schedule and cost of the previous plan. It must be clear that this investment in risk mitigation is not a guarantee of being risk free, but it does put in place guards against downside risk. Residual risks R5 (Clinical Validation Bottleneck) and R7 (Team Knowledge Loss) remain accepted without dedicated mitigation tasks and will be monitored at each review cycle.

Control and backup: If “D5 Full mitigation” is infeasible due to budget requirements the backup suggestion would be “D3 Clinical validation”. As we transition from our previous state to lower risk states, we see a large cost shift (see fig 3.1) when we add clinical review mitigations against hallucinations. We view this mitigation as a key requirement to ensure the likelihood of model hallucinations do not impede proper clinical care. Our backup plan does have some cost reduction compared to our new default of D5, but it still exceeds our budgetary target set. For control there should be quarterly reviews and re-forecasting. This allows for any breakthroughs in model quality or shifts in technical landscapes to be augmented and accommodated. Re-planning — beyond re-forecasting — is triggered when a risk event materializes, two or more sequential tasks slip, or a scope change is requested. Key assumptions for this forecast to remain feasible: Mayo Clinic IT provides lab data access within weeks 1–6, IRB pre-submission initiated in week 1 completes within 10 weeks, and physician reviewers are available for clinical validation sessions during weeks 30–38. Quarterly reviews should include the project team and Mayo Clinic stakeholders jointly, where the project manager presents the updated cost × riskiness tradespace. The Mayo Clinic sponsor owns the decision to re-plan, while the project manager owns re-forecasting.

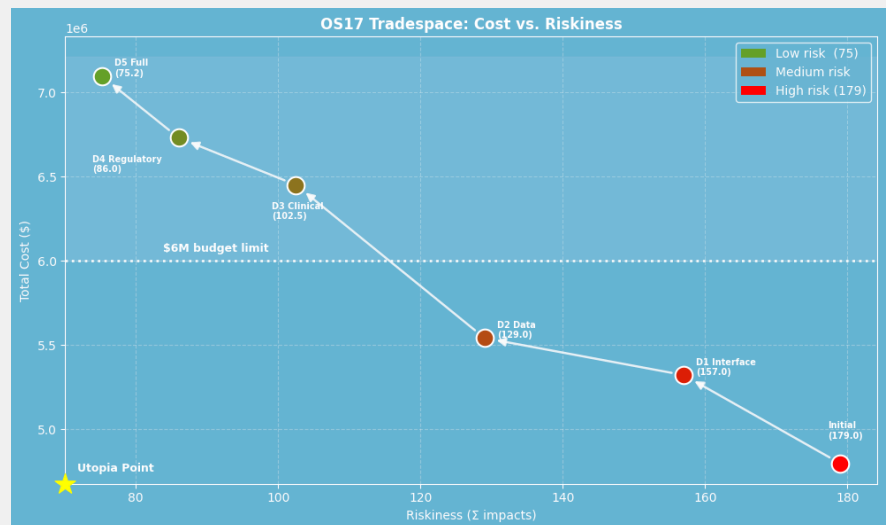


Fig 3.1 - Tradespace of risk versus cost for design decisions

Appendix 1: List of Mitigations

ID	Risk	Mitigation	Details	New Effect	New Prob.	New Impact	Commentary
M1	R1	Data Integration	6 person-weeks; DATA + CLOUD; runs in parallel with T1.1	35	20%	7	Early technical proof-of-concept with Mayo Clinic IT. Reduces probability from 50% to 20% by discovering integration blockers before they cascade. Effect reduces because we have time to pivot architecture.
M2	R2	Clinical Validation Readiness	8 person-weeks; AI + QA; starts after T2.2	50	15%	7.5	Structured clinical validation with a physician review board run before shadow mode. Reduces probability from 40% to 15%; residual effect remains high (50) because hallucination can never be fully eliminated from LLM-based systems.
M3	R3	Regulatory Readiness	4 person-weeks; PM + QA; runs in parallel with T1.1	30	25%	7.5	Early IRB pre-submission and regulatory roadmap during the design phase. Reduces the probability from 40% to 25% and the effect from 60% to 30% by clarifying what evidence, documentation, and clinical study framing are needed before T3.2 (Shadow Mode & Clinical Review). This reduces the chance of discovering regulatory gaps only at the validation gate.
M4	R4	PHI Compliance Control	4 person-weeks; CLOUD + DATA + QA; starts after T1.3	55	5%	2.75	Focused review of PHI handling, access control, logging, audit trail, data minimization, and compliance assumptions before real patient data is used in validation. Reduces probability from 15% to 5% by finding privacy and security design issues before T3.2 (Exception & Resilience Testing) and T3.3 (Shadow-Mode Clinical Validation). The effect is reduced from 90% to 55% because the early review allows the team to redesign and implement new controls before the issues affect the project as a whole.
M6	R6	Early Integration & Interface Testing	6 person-weeks; SWE + DATA + AI + QA; starts after T2.1	40	20%	8	Early interface review and integration testing across the data pipeline, retrieval layer, agentic AI core, physician-facing workflow, and validation workflow. Reduces the probability from 40% to 20% by exposing interface assumptions before they become late problems. The effect is reduced from 75% to 40% because the difficulties and special integration needs identified during initial testing are easier to address than the cascading rework observed in phase T3.3 (Shadow-Mode Clinical Validation).

Appendix 2: Costs of Mitigations

The person-weeks describe the internal team effort required to execute each mitigation task. Since the baseline forecast already includes the full-time team burn rate, the extra costs mentioned were modeled as incremental mitigation costs for resources not fully captured by internal staffing, such as clinical expert time, regulatory support, compliance review, cloud testing, data access, sandbox setup, and early integration environments.

Rates are sourced from the BLS Occupational Outlook Handbook (2025) [1]-[6] and reflect median US market compensation (same as used in OS16).

ID	Mitigation	Internal Effort	Cost	Rationale
M1	Data Integration	25 person-weeks	\$220,000.00	Work shared across DATA, CLOUD, and SWE to set up a sandbox environment, obtain representative lab data, test cloud access, map EHR/Lab data formats, and validate early harmonization assumptions with Mayo Clinic IT.
M2	Clinical Validation Readiness	65 person-weeks	\$700,000.00	Cross-functional effort involving AI, QA, PM, and clinical reviewers. This is the highest extra-cost mitigation because it depends on limited physician availability. The effort covers validation-case preparation, review protocol definition, physician panel sessions, output review, and clinical validation material.
M3	Regulatory Readiness	30 person-weeks	\$280,000.00	Work from PM and QA, with input from clinical and compliance stakeholders. It covers IRB pre-submission preparation, study framing, documentation review, evidence planning, and alignment of the validation plan with institutional approval expectations.
M4	PHI Compliance Control	35 person-weeks	\$260,000.00	Combined CLOUD, DATA, QA, and compliance support effort. It covers PHI handling review, access control, audit trail, logging, encryption assumptions, minimum-necessary data use, and updates to compliance controls before patient data is used in validation.
M5	Early Integration & Interface Testing	50 person-weeks	\$420,000.00	Work across SWE, DATA, AI, and QA. It covers early integration setup, cross-team review sessions, interface testing, contract alignment, and end-to-end checks across the data pipeline, retrieval layer, agentic AI core, physician-facing workflow, and clinical validation workflow.

ID	Risk	Mitigation	Details	New Effect	New Prob.	New Impact	Commentary
M1	R1	Data Integration Feasibility Spike	T_M1: 6 pw, DATA+CLOUD teams; unblocked, runs	35	20%	7.0	Early technical proof-of-concept with Mayo Clinic IT. Reduces probability from 50% → 20% by discovering integration blockers before they cascade. Effect reduces because we have time to pivot architecture.

ID	Risk	Mitigation	Details	New Effect	New Prob.	New Impact	Commentary
			concurrently with T1.1				
M2	R2	Clinical Validation Protocol & Physician Expert Panel	T_M2: 8 pw, AI+QA teams; blocked by T2.2 (Agentic Core)	50	15%	7.5	Structured clinical validation with a physician review board run before shadow mode. Reduces probability from 40% → 15%; residual effect remains high (50) because hallucination can never be fully eliminated from LLM-based systems.
M3	R3	IRB Pre-submission & Regulatory Roadmap	T_M3: 4 pw, PM team; unblocked, runs concurrently with T1.1	30	25%	7.5	Engage regulatory consultant and initiate IRB pre-submission in Design phase (parallel with T1.1). Reduces probability from 55% → 25% and effect from 60 → 30 by building regulatory alignment early and avoiding surprises at Shadow Mode gate.

Appendix 3: Project Model

We continued using the same project model, developed from scratch in Python in Google Colab, and extended it to incorporate the risk mapping required for OS17. The model is based on an agent-based framework centered on Teams and Tasks, and it also includes Phases, Products, Productivity multipliers (such as overtime), and explicit risk elements. These additions allowed us to connect the original OS16 project simulation with the OS17 risk-mitigation analysis. We wrote the code by hand with extensive comments and minimal AI assistance. The model remains publicly viewable in the [🔗 OS16 Colab](#) and is detailed in Appendix 2: Model Description of OS16.

Appendix 4: Citations and AI Disclosure

Citations:

(note: these citations are from OS16 and formed the basis for our model generation that we build upon for OS17)

- [1] U.S. Bureau of Labor Statistics, “Software Developers, Quality Assurance Analysts, and Testers,” Occupational Outlook Handbook, 2025. [Online]. Available: <https://www.bls.gov/ooh/computer-and-information-technology/software-developers.htm>. [Accessed: Apr. 12, 2026].
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- [5] U.S. Bureau of Labor Statistics, “Database Administrators and Architects,” Occupational Outlook Handbook, 2025. [Online]. Available: <https://www.bls.gov/ooh/computer-and-information-technology/database-administrators.htm>. [Accessed: Apr. 12, 2026].

[6] U.S. Bureau of Labor Statistics, "Project Management Specialists," Occupational Outlook Handbook, 2025. [Online]. Available: <https://www.bls.gov/ooh/business-and-financial/project-management-specialists.htm>. [Accessed: Apr. 12, 2026].

AI Disclosure:

ChatGPT to support the development of this assignment by reviewing text clarity, improving wording, organizing the mitigation analysis, and helping interpret the cost, schedule, and risk tradespace results. AI assistance was also used to refine descriptions of risks, mitigations, design walk logic, and figure commentary. All project assumptions, model structure, numerical results, and final decisions were reviewed and edited by the team.

AI was also used to aid generating code and interpretation of code written by other team members, but the vast majority of the code and tests were written by hand. Google's Gemini was used to review drafts of this assignment and act as the role of a MITsdm TA providing feedback on sections with strong arguments and areas requiring improvement. All work in response to this feedback along with all modeling choices, inputs, outputs, and conclusions are the team's own.